**EXPERIMENT 16: Frequency Distribution of Words in Customer Reviews**

**1. AIM**

To develop a Python program to calculate the frequency distribution of words in a customer reviews dataset.

**2. ALGORITHM**

1. Import the Counter class from the collections module.
2. Read the customer reviews from a text file.
3. Convert all words to lowercase.
4. Remove punctuation marks and split the text into individual words.
5. Count the frequency of each word using Counter.
6. Display the words and their frequencies.

**3. PYTHON CODE**

Experiment 16 – Python Code

```
from collections import Counter
import string

# Read customer reviews from a text file
with open("customer_reviews.txt", "r") as file:
    reviews = file.read().lower()

# Remove punctuation
reviews = reviews.translate(str.maketrans("", "", string.punctuation))

# Split reviews into words
words = reviews.split()

# Calculate word frequency
frequency = Counter(words)
```

```
# Display the frequency distribution

print("Frequency Distribution of Words:")

for word, count in frequency.items():

   print(word, ":", count)
```

**4. INPUT**

Create a text file named **customer_reviews.txt** in the same folder as your Python program.

Enter the following sample reviews into the file:

The product is good.

The product is useful.

Good product and good quality.

**Important:** Save the text file in the same folder as your Python program before running the code.

**5. OUTPUT**

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 16.py ========
Frequency Distribution of Words:
the : 2
product : 3
is : 2
good : 3
useful : 1
and : 1
```

**6. RESULT**

Thus, the Python program to calculate and display the frequency distribution of words in customer reviews was executed successfully.